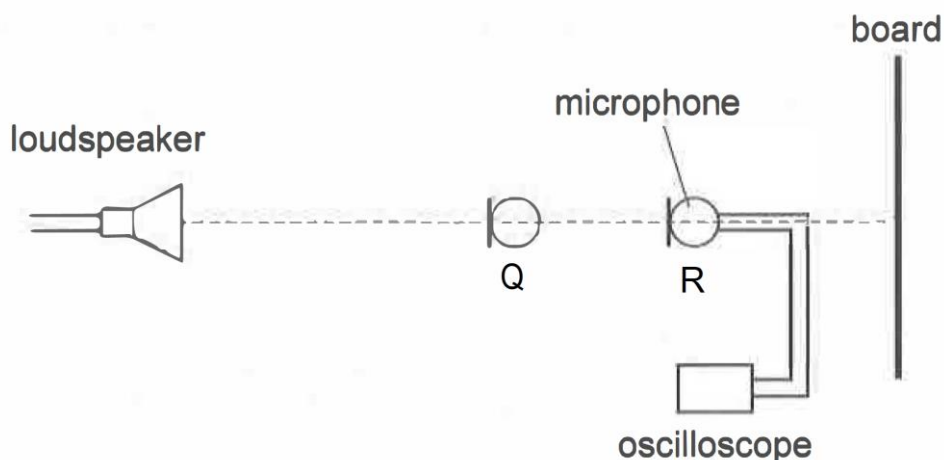
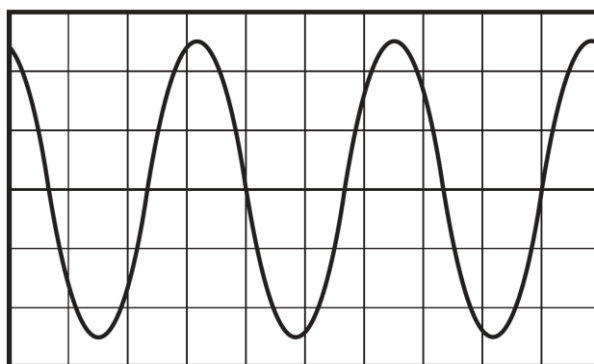


18 Stationary sound waves are investigated using the apparatus shown.



Q and R are adjacent positions of the microphone that give maximum amplitude on the oscilloscope, as shown.



oscilloscope screen when microphone is at R

The distance QR is 6.0 cm and the time base of the oscilloscope is $20 \mu\text{s div}^{-1}$. What are the frequency and the wavelength of the sound wave?

	frequency / kHz	wavelength / cm	
A	15	6.0	
B	15	12	
C	20	6.0	
D	20	12	

Answer: B

Distance between two antinodes = $\frac{\lambda}{2} = 6.0 \text{ cm}$

Wavelength = $6.0 \times 2 = 12 \text{ cm}$

1.5 T as seen from the oscilloscope = $5 \times 20 \times 10^{-6}$

$T = \frac{5 \times 20 \times 10^{-6}}{1.5}$

Frequency = $\frac{1}{T} = \frac{1.5}{5 \times 20 \times 10^{-6}} = 15\,000 \text{ Hz} = 15 \text{ kHz}$

10	Monochromatic light is incident on a pair of narrow slits a distance of 0.1 mm apart. A series
----	--